A cone and compactness argument for statistical instability of a stochastically forced shear

Reviewed research draft

22 October 2026

Abstract

For a fixed elongated rectangular torus and forcing in the single vorticity mode $\sin y$, we give an argument for existence, at sufficiently small viscosity, of an ergodic invariant measure not supported on shear flows. The key is an exact energy–enstrophy identity. It provides tightness of a normalized transverse process even if its amplitude tends to zero. Concentration on the shear law would then produce an invariant probability on a negative-energy hyperboloid for the frozen linearized Euler equation. A Fourier-block argument rules out that probability. The argument targets existence of a non-shear stationary state; it does not estimate a Lyapunov exponent or an explicit viscosity threshold.

1 Statement and normalization

Fix $0 < \varepsilon < 1$ and

$$\mathbb{T}^2_\varepsilon = (\mathbb{R}/(2\pi\varepsilon^{-1}\mathbb{Z})) \times (\mathbb{R}/(2\pi\mathbb{Z})).$$

Use normalized Lebesgue measure, let $H = L^2_0(\mathbb{T}^2_\varepsilon;\mathbb{R})$, and set $A = -\Delta$. Its eigenvalues are $\lambda_{m,n} = \varepsilon^2 m^2 + n^2$, excluding $(m,n) = (0,0)$. All velocities have zero spatial mean. With $\nabla^\perp = (-\partial_y, \partial_x)$, put

$$u(w) = -\nabla^\perp A^{-1} w, \qquad \mathcal{B}(v,w) = u(v)\cdot\nabla w, \qquad e(x,y) = \sin y.$$

Consider

$$d\omega = (-\nu A\omega - \mathcal{B}(\omega,\omega))\,dt + \sigma e\,dW_t, \qquad \nu,\sigma > 0. \tag{1}$$

Let $\mathcal{S} = \{\omega : \partial_x\omega = 0\}$ and $P_x\omega$ be the horizontal average.

Theorem 1.1. *For every fixed $\varepsilon \in (0,1)$ there exists $\eta_0(\varepsilon) > 0$ such that, whenever $\nu^{3/2}/\sigma < \eta_0(\varepsilon)$, equation (1) has an ergodic invariant probability μ_* with $\mu_*(\mathcal{S}) = 0$. In fact it can be chosen to satisfy*

$$\mu_*\left(\|u(\omega)\|_2^2 - \|\omega\|_2^2 > 0\right) = 1.$$

Consequently, for $F(\omega) = \min\{1, \|(I-P_x)\omega\|_2^2\}$, there exist deterministic non-shear initial conditions for which

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T F(\omega_t)\,dt = \int F\,d\mu_* > 0 \qquad \textit{almost surely.}$$

The Gaussian invariant shear law therefore is not the unique invariant law.

The normalization

$$\tau = \frac{\sigma}{\sqrt{\nu}}t, \qquad w_\tau = \frac{\sqrt{\nu}}{\sigma}\omega_{\sqrt{\nu}\tau/\sigma}, \qquad \eta = \frac{\nu^{3/2}}{\sigma}$$

converts (1) into

$$dw = (-\eta A w - \mathcal{B}(w,w))\,d\tau + \sqrt{\eta}\,e\,dB_\tau. \tag{2}$$

The Brownian scaling is included in this calculation. It suffices to prove the theorem for (2) as $\eta \downarrow 0$. In the rest of the proof the time variable is again denoted by t.

For each fixed $\eta > 0$, we use the usual global mean-zero L^2 vorticity solution in two dimensions, its continuous dependence on initial data, and its Feller Markov semigroup; see the framework in [1]. The noise is smooth and finite rank. The energy calculations below are justified first for smooth approximations and then by parabolic regularization and the L^2 energy bounds. No irreducibility or uniqueness theorem for the full process is assumed.

2 An exact identity and an invariant cone

Define

$$\mathcal{K}(w) = \langle (A^{-1} - I)w, w\rangle = \|u(w)\|_2^2 - \|w\|_2^2, \tag{3}$$

$$\mathcal{D}(w) = \|(A - I)A^{-1/2}w\|_2^2 = \sum_{\lambda} \frac{(\lambda - 1)^2}{\lambda} |w_\lambda|^2. \tag{4}$$

The form $\mathcal{K}$ is continuous on H. Extend $\mathcal{D}$ to $+\infty$ when its series diverges.

Lemma 2.1. *Along* (2),

$$\frac{d}{dt}\mathcal{K}(w_t) = 2\eta\{\mathcal{D}(w_t) - \mathcal{K}(w_t)\} \tag{5}$$

in integrated form. In particular $\mathcal{K}(w_t) \geq e^{-2\eta t}\mathcal{K}(w_0)$ *whenever* $\mathcal{K}(w_0) > 0$.

Proof. Both kinetic energy and enstrophy are conserved by the Euler nonlinearity:

$$\langle A^{-1}w, \mathcal{B}(w,w)\rangle = \langle w, \mathcal{B}(w,w)\rangle = 0.$$

Since $Ae = e$, both the martingale term and the Itô correction in $d\mathcal{K}$ vanish. The viscous contribution is

$$2\langle (A^{-1} - I)w, -\eta A w\rangle = 2\eta\langle (A - I)w, w\rangle = 2\eta(\mathcal{D}(w) - \mathcal{K}(w)).$$

Finally $\mathcal{D} \geq 0$, which gives the claimed inequality. □

The cone $\{\mathcal{K} > 0\}$ is nonempty: $w_0 = \cos(\varepsilon x)$ lies in it. Every shear has $\mathcal{K} \leq 0$, since its nonzero Fourier eigenvalues are $n^2 \geq 1$. Thus $\{\mathcal{K} > 0\}$ is disjoint from $\mathcal{S}$.

Let $E = \ker(A - I)$, let P be its orthogonal projection, and put $Q = I - P$. We remove the *whole* eigenspace E, including horizontal modes if ε^{-1} is an integer. This also handles resonant aspect ratios. Since the spectrum is discrete, there is a constant $c_\varepsilon > 0$ such that

$$\mathcal{D}(q) \geq c_\varepsilon \|q\|_{H^1}^2, \qquad q \in QH. \tag{6}$$

Indeed $(\lambda - 1)^2/[\lambda(1 + \lambda)]$ has a positive infimum over eigenvalues $\lambda \neq 1$.

3 Retaining the direction when the amplitude vanishes

For a trajectory starting with $\mathcal{K}(w_0) > 0$, introduce

$$a = Pw, \qquad \rho = \sqrt{\mathcal{K}(w)}, \qquad \theta = \frac{Qw}{\rho}.$$

Then $\mathcal{K}(\theta) = 1$, $\mathcal{D}(w) = \rho^2 \mathcal{D}(\theta)$, and $w = a + \rho\theta$. The lifted state space is the Polish space

$$\mathcal{X} = E \times [0, \infty) \times \mathcal{M}, \qquad \mathcal{M} = \{\theta \in QH : \mathcal{K}(\theta) = 1\},$$

with its finite-dimensional times strong-L^2 topology. Let $\mathcal{X}_0 = \{\rho = 0\}$.

For $\rho > 0$, the exact equations are

$$da = [-\eta a - \rho P\{\mathcal{B}(a, \theta) + \mathcal{B}(\theta, a)\} - \rho^2 P\mathcal{B}(\theta, \theta)]\, dt + \sqrt{\eta}\, e\, dB_t, \tag{7}$$

$$\dot{\rho} = \eta\{\mathcal{D}(\theta) - 1\}\rho, \tag{8}$$

$$\dot{\theta} = -\eta A\theta - Q\{\mathcal{B}(a, \theta) + \mathcal{B}(\theta, a)\} - \rho Q\mathcal{B}(\theta, \theta) - \eta\{\mathcal{D}(\theta) - 1\}\theta. \tag{9}$$

Here $\mathcal{B}(a, a) = 0$ for every $a \in E$, because $A^{-1}a = a$. Equations involving $\mathcal{D}$ are interpreted in integrated/mild form.

Lemma 3.1. *The lifted dynamics extend to a global Feller Markov process on $\mathcal{X}$. On $\mathcal{X}_0$, a solves*

$$da = -\eta a\, dt + \sqrt{\eta}\, e\, dB_t.$$

Given this a, let

$$\dot{z} = -\eta A z - Q\{\mathcal{B}(a, z) + \mathcal{B}(z, a)\}, \qquad z_0 = \theta_0. \tag{10}$$

The boundary direction is $\theta_t = z_t/\sqrt{\mathcal{K}(z_t)}$.

Proof. The quadratic identity for (10) is again $\dot{\mathcal{K}}(z) = 2\eta(\mathcal{D}(z) - \mathcal{K}(z))$. One way to check the nonlinear cancellation is to expand the identity $\langle (A^{-1} - I)w, \mathcal{B}(w, w)\rangle = 0$ at $w = a + sz$: $(A^{-1} - I)a = 0$ and $\mathcal{B}(a, a) = 0$, so its coefficient of s^2 is precisely the required cancellation. Thus $\mathcal{K}(z_t) \geq e^{-2\eta t} > 0$, and the boundary formula is global.

We give details of continuity at $\rho_0 = 0$, where division by a vanishing physical amplitude would otherwise be problematic. For an initial amplitude $r \geq 0$, solve instead for (a, v) with $w = a + rv$:

$$da = [-\eta a - rP\{\mathcal{B}(a, v) + \mathcal{B}(v, a)\} - r^2 P\mathcal{B}(v, v)]\, dt + \sqrt{\eta}\, e\, dB_t,$$
$$\dot{v} = -\eta A v - Q\{\mathcal{B}(a, v) + \mathcal{B}(v, a)\} - rQ\mathcal{B}(v, v), \qquad v_0 = \theta_0.$$

For $r > 0$ these are the physical solution written in rescaled coordinates. For $r = 0$ they are the boundary linear system. The identity gives $\mathcal{K}(v_t) \geq e^{-2\eta t}$ in every case. The lifted variables are $\rho_t = r\sqrt{\mathcal{K}(v_t)}$ and $\theta_t = v_t/\sqrt{\mathcal{K}(v_t)}$.

On every finite time interval, bounded initial data and a bounded realization of the smooth additive-noise path give a uniform physical L^2 bound, hence a uniform bound on a in every spatial norm (as E is finite dimensional). In the v energy estimate both $\mathcal{B}(a, v)$ and $r\mathcal{B}(v, v)$ cancel:

$$\frac{1}{2}\frac{d}{dt}\|v\|_2^2 + \eta\|A^{1/2}v\|_2^2 = -\langle \mathcal{B}(v, a), v\rangle \leq C_\varepsilon \|a\|_{C^1}\|v\|_2^2.$$

These estimates bound v uniformly in $C([0, T]; L^2) \cap L^2(0, T; H^1)$ for initial lifted states in a bounded neighborhood, with $\eta > 0$ fixed. Here and below the constants may depend on η, T, ε and the fixed noise path; no uniformity as $\eta \to 0$ is needed in this lemma.

To check continuity at a boundary state, take $(a_{0,j}, r_j, \theta_{0,j}) \to (a_0, 0, \theta_0)$ and drive all solutions with the same Brownian motion. Physical continuous dependence gives

$$w_j \longrightarrow a^0 \quad \text{in } C([0, T]; L^2), \qquad a_j = Pw_j \longrightarrow a^0 \quad \text{in } C([0, T]; C^1),$$

where a^0 is the boundary Ornstein–Uhlenbeck solution from a_0. For indices with $r_j = 0$, the same statement follows directly from the finite-dimensional equation. Let z solve (10) with a^0 and initial value θ_0. For $d_j = v_j - z$, the difference equation is

$$\dot{d}_j = -\eta A d_j - Q\{\mathcal{B}(a_j, d_j) + \mathcal{B}(d_j, a_j)\} + R_j,$$
$$R_j = -Q\{\mathcal{B}(a_j - a^0, z) + \mathcal{B}(z, a_j - a^0)\} - r_j Q\mathcal{B}(v_j, v_j).$$

The required forcing convergence is explicit. In two dimensions,

$$\|\mathcal{B}(v,v)\|_{H^{-1}} \le C_\varepsilon \|u(v)\|_{L^4}\|v\|_{L^4} \le C_\varepsilon \|v\|_2^{3/2}\|v\|_{H^1}^{1/2},$$
$$\|\mathcal{B}(b,z)+\mathcal{B}(z,b)\|_{H^{-1}} \le C_\varepsilon \|b\|_{C^1}\|z\|_2, \qquad b \in E.$$

Consequently $R_j \to 0$ in $L^2(0,T;H^{-1})$. For the quadratic term, this follows from

$$\int_0^T \|r_j Q\mathcal{B}(v_j,v_j)\|_{H^{-1}}^2\,dt \le C_\varepsilon r_j^2 \sup_{t\le T}\|v_j(t)\|_2^3\, T^{1/2}\left(\int_0^T \|v_j(t)\|_{H^1}^2\,dt\right)^{1/2}.$$

Taking the L^2 energy estimate for d_j, absorbing the forcing with the viscous term, and using Gronwall's inequality gives

$$\sup_{t\le T}\|d_j(t)\|_2^2 \le C_{\eta,T}\left(\|\theta_{0,j}-\theta_0\|_2^2 + \|R_j\|^2_{L^2(0,T;H^{-1})}\right) \longrightarrow 0.$$

The bound $\mathcal{K}(v_j(t)) \ge e^{-2\eta T}$ ensures continuity of normalization. The amplitudes $r_j\sqrt{\mathcal{K}(v_j(t))}$ also tend to zero uniformly on $[0,T]$. Interior continuity follows directly from physical continuous dependence and the same positive lower bound for $\mathcal{K}(w_t)$. Thus the lifted solutions depend continuously on initial data, almost surely under this coupling. Dominated convergence for bounded continuous test functions proves the Feller property. The boundary linear flow and its homogeneous normalization satisfy the flow property, so this extension is indeed Markov. Global existence in the interior follows from global two-dimensional Navier–Stokes; at the boundary it follows from the linear equation. □

The reconstruction map $\pi(a,\rho,\theta) = a + \rho\theta$ is continuous and intertwines the lifted and physical semigroups, including at $\rho = 0$.

4 An invariant lifted law with a viscosity-independent directional bound

Proposition 4.1. *For every $\eta > 0$ the lifted process has an invariant probability $\widehat{\mu}_\eta$ such that*

$$\int_{\mathcal{X}} \mathcal{D}(\theta)\,d\widehat{\mu}_\eta \le 1. \tag{11}$$

Proof. Start from a fixed smooth w_0 with $\mathcal{K}(w_0) > 0$. The enstrophy identity yields

$$\mathbb{E}\|w_T\|_2^2 + 2\eta\int_0^T \mathbb{E}\|A^{1/2}w_t\|_2^2\,dt = \|w_0\|_2^2 + \eta T\|e\|_2^2.$$

Poincaré's inequality also gives $\sup_{t\ge 0}\mathbb{E}\|w_t\|_2^2 < \infty$ for fixed initial data, uniformly in $\eta > 0$. Since $\mathcal{K}(w) \le C_\varepsilon\|w\|_2^2$, Jensen's inequality gives

$$\mathbb{E}\log\mathcal{K}(w_T) \le \log\mathbb{E}\mathcal{K}(w_T) \le C$$

with a finite T-independent upper bound. The lower bound $\log\mathcal{K}(w_T) \ge \log\mathcal{K}(w_0) - 2\eta T$ ensures integrability. By Lemma 2.1,

$$\log\mathcal{K}(w_T) - \log\mathcal{K}(w_0) = 2\eta\int_0^T \{\mathcal{D}(\theta_t) - 1\}\,dt.$$

Taking expectations therefore gives

$$\limsup_{T\to\infty}\frac{1}{T}\int_0^T \mathbb{E}\mathcal{D}(\theta_t)\,dt \le 1. \tag{12}$$

The enstrophy bound controls the time averages of $\|a\|_2^2+\rho^2$. Equation (12) and (6) control the time average of $\|\theta\|_{H^1}^2$. The compact embedding $H^1 \Subset L^2$, together with finite dimensionality of E and of ρ, makes the expected occupation measures on $\mathcal{X}$ tight. Their weak limits are invariant by the Feller property and the Krylov–Bogoliubov argument. Lower semicontinuity of $\mathcal{D}$ proves (11). □

The bound is on the *direction after division by the amplitude*. This is the additional compactness information that ordinary physical energy estimates alone would not supply.

5 The frozen Euler equation has no invariant law on $\mathcal{M}$

On the boundary, the only invariant law of a is

$$a = fe, \qquad f \sim \gamma := N(0, 1/2).$$

All unforced coordinates in E decay exponentially. For $a = f \sin y$ define

$$B = I - A^{-1}, \qquad L = -Q \cos y \, \partial_x B.$$

The inviscid boundary flow with f frozen is

$$(f, \theta) \longmapsto (f, e^{tfL}\theta).$$

To construct the group, extend the displayed operator to H and write

$$L = -\cos y \, \partial_x + P \cos y \, \partial_x + Q \cos y \, \partial_x A^{-1}.$$

The first term generates a transport group. The second extends to a bounded finite-rank operator, by integration by parts against the smooth basis of E, and the third is bounded on L^2. Bounded perturbation therefore gives a strongly continuous group, which preserves QH since $PL = 0$. It conserves $\mathcal{K} = -\langle B\cdot, \cdot\rangle$, first on smooth data and then by density. Complex inner products below are linear in their first argument.

Lemma 5.1. *For every horizontal wavenumber $0 < k < 1$, the operator*

$$L_k = -ik\, C B_k, \qquad C = \cos y, \quad B_k = I - (k^2 - \partial_y^2)^{-1}$$

on complex $L^2(\mathbb{T}_y)$ has eigenvectors v_+, v_- with eigenvalues $+\lambda_k, -\lambda_k$, where $\lambda_k > 0$. Their span has one positive and one negative direction for the Hermitian form $\langle B_k\cdot, \cdot\rangle$. Its B_k-orthogonal complement is nonnegative for that form.

Proof. The diagonal Fourier entries of B_k are $b_n = 1 - (k^2 + n^2)^{-1}$. Only b_0 is negative; all other entries are strictly positive. Thus B_k has negative index one and is boundedly invertible.

Split functions into Fourier modes with even and odd integer n, denoting the two spaces by H_e, H_o. Multiplication by $\cos y$ interchanges them. Write B_e, B_o for the restrictions of B_k and $C_{oe} : H_e \to H_o$, $C_{eo} = C_{oe}^*$ for the two parts of C. The operator

$$G = C_{eo} B_o C_{oe}$$

is bounded, self-adjoint, nonnegative and injective: B_o is strictly positive and multiplication by $\cos y$ has no nonzero L^2 kernel. Hence $\operatorname{Ran} G^{1/2}$ is dense in H_e.

Consider the self-adjoint operator

$$T = G^{1/2} B_e G^{1/2}.$$

It is a compact perturbation of G, because $I - B_e$ is compact. It has a negative direction, since $\operatorname{Ran} G^{1/2}$ is dense and B_e is negative on the constant Fourier mode. Consequently T has a negative isolated eigenvalue $-a_k^2$, $a_k > 0$. Its negative index is at most one, so this eigenvalue is simple. Choose u with $Tu = -a_k^2 u$ and put

$$p = G^{1/2}u, \qquad \lambda_k = ka_k, \qquad q = -\frac{ik}{\lambda_k} C_{oe} B_e p.$$

Then $GB_e p = -a_k^2 p$, so direct substitution gives $L_k(p,q) = \lambda_k(p,q)$ and $L_k(p,-q) = -\lambda_k(p,-q)$. Thus take $v_\pm = (p, \pm q)$.

Writing $h = \langle B_e p, p\rangle = -a_k^2 \|u\|^2 < 0$, we obtain

$$\langle B_o q, q\rangle = -h, \qquad \langle B_k v_\pm, v_\pm\rangle = 0, \qquad \langle B_k v_+, v_-\rangle = 2h \neq 0.$$

The span has signature $(1,1)$. Its orthogonal complement cannot have a negative direction, since that would give two negative directions for a form whose negative index is one. □

Proposition 5.2. *There is no invariant probability for $(f,\theta) \mapsto (f, e^{tfL}\theta)$ on $\mathbb{R} \times \mathcal{M}$ whose f-marginal is γ.*

Proof. Disintegrate such a probability with respect to f. For γ-almost every $f \neq 0$, the conditional law of θ is invariant under e^{tfL} (it suffices to use the time-one map).

For every positive horizontal wavenumber $k = m\varepsilon < 1$, use the eigenvectors from Lemma 5.1. The bounded linear observables

$$\ell_+(z) = \langle B_k z, v_-\rangle, \qquad \ell_-(z) = \langle B_k z, v_+\rangle$$

are multiplied by $e^{tf\lambda_k}$ and $e^{-tf\lambda_k}$, respectively. Here we use $L_k^* B_k + B_k L_k = 0$. A probability on $\mathbb{C}$ invariant under multiplication by a positive real constant different from one is concentrated at zero: after replacing that constant by its reciprocal if necessary, it is $c > 1$; the disjoint annuli $\{c^j \le |z| < c^{j+1}\}$, $j \in \mathbb{Z}$, all have the same mass, which must be zero, and cover $\mathbb{C} \setminus \{0\}$. Thus both observables vanish almost surely in every such block. Lemma 5.1 implies $\langle B_k\theta_k, \theta_k\rangle \ge 0$ there.

For $|k| \ge 1$ the form is already nonnegative. At $k = 0$ it is nonnegative after removing the mean and the $A = 1$ eigenspace. The projection Q does not affect any block with $0 < |k| < 1$, since that block contains no $A = 1$ eigenvalue. Summing all horizontal blocks, with the usual reality relation for negative k, yields $\langle B\theta, \theta\rangle \ge 0$. This contradicts $\mathcal{K}(\theta) = -\langle B\theta, \theta\rangle = 1$. □

6 Passing invariant boundary laws to zero viscosity

Lemma 6.1. *There cannot exist a sequence $\eta_j \downarrow 0$ of invariant probabilities ζ_j for the boundary lifted process satisfying $\int \mathcal{D}(\theta)\, d\zeta_j \le 1$.*

Proof. Under any stationary boundary law, $a = fe$ and

$$df = -\eta_j f\, dt + \sqrt{\eta_j}\, dB_t, \qquad \operatorname{Law}(f_t) = \gamma.$$

The directional bound and (6) make the joint laws of (f,θ) tight in $\mathbb{R} \times QH$. Every weak limit is supported on $\mathbb{R} \times \mathcal{M}$ and has marginal γ.

We verify that a weak limit is invariant for the frozen Euler flow. The unnormalized boundary equation is

$$\dot z_j = -\eta_j A z_j + f_j(t) L z_j, \qquad z_j(0) = \theta.$$

For a fixed finite time interval, $f_j(t) \to f(0)$ uniformly in probability, uniformly for initial $f(0)$ in compact sets. The energy estimate

$$\frac{d}{dt}\|z_j\|_2^2 \le C_\varepsilon |f_j(t)| \|z_j\|_2^2$$

is uniform in η_j. For smooth initial data put $z_0(t) = e^{tf(0)L}\theta$. The equation for $d_j = z_j - z_0$ has homogeneous part $-\eta_j A + f_j(t)L$ and forcing $-\eta_j A z_0 + (f_j - f(0))Lz_0$. On the event $\sup_{t\le T}|f_j(t)| \le R$, its L^2 estimate gives

$$\sup_{t\le T}\|d_j(t)\|_2 \le C_{R,T,\varepsilon}\left[\eta_j \int_0^T \|Az_0(t)\|_2\,dt + \sup_{t\le T}|f_j(t) - f(0)| \int_0^T \|Lz_0(t)\|_2\,dt\right]. \tag{13}$$

This estimate uses L^2 forcing rather than absorption into viscosity, so its constant does not contain η_j^{-1}. Transport by the smooth vector field, together with the bounded smoothing and finite-rank terms, preserves H^2 over finite time intervals; the integrals on the right are therefore finite and bounded uniformly for $f(0)$ in compact sets and smooth initial data in a fixed bounded set.

Approximation by smooth data and the uniform energy bound extend convergence to arbitrary L^2 initial data. A finite smooth net of any compact set of initial directions makes this convergence uniform on that set. For initial $|f(0)| \le R_0$, the explicit Ornstein–Uhlenbeck formula and a martingale maximal estimate give

$$\mathbb{E}\sup_{t\le T}|f_j(t) - f(0)|^2 \le C_{T,R_0}(\eta_j + \eta_j^2) \longrightarrow 0.$$

Thus one may take any fixed $R > R_0$ above, with complementary event of probability tending to zero. This proves convergence uniformly in probability over compact sets of initial data in $\mathbb{R} \times QH$.

Since $\mathcal{K}(z_0(t)) = 1$, continuity of $\mathcal{K}$ gives

$$\frac{z_j(t)}{\sqrt{\mathcal{K}(z_j(t))}} \longrightarrow e^{tf(0)L}\theta$$

in probability, again uniformly on compact subsets of $\mathbb{R} \times \mathcal{M}$. The denominator is always defined by Lemma 3.1. Write P_j^t for the boundary semigroup and $\mathsf{P}_0^t\varphi(f,\theta) = \varphi(f, e^{tfL}\theta)$. For every bounded continuous φ, the established convergence gives $\mathsf{P}_j^t\varphi \to \mathsf{P}_0^t\varphi$ uniformly on compact sets. Indeed, the limiting image of a compact set is compact, and continuity of φ controls its values in a sufficiently small neighborhood of that image. Tightness, weak convergence, and invariance now give

$$\int \varphi\,d\zeta = \lim_j \int \mathsf{P}_j^t\varphi\,d\zeta_j = \int \mathsf{P}_0^t\varphi\,d\zeta.$$

The weak limit is therefore invariant for the frozen Euler flow. Proposition 5.2 gives a contradiction. □

7 Completion of the nonlinear argument

Lemma 6.1 implies that there is an $\eta_0 > 0$ such that, for $0 < \eta < \eta_0$, no invariant law supported on $\mathcal{X}_0$ can satisfy $\int \mathcal{D} \le 1$. Take the invariant lifted law $\widehat{\mu}_\eta$ furnished by Proposition 4.1. It must give positive mass to $\{\rho > 0\}$.

Both $\{\rho = 0\}$ and $\{\rho > 0\}$ are invariant for the lifted process. The latter follows from the strict lower bound on $\mathcal{K}$; the former follows from the definition of the boundary

extension. Conditioning $\widehat{\mu}_\eta$ on $\{\rho > 0\}$ therefore yields an invariant probability. Reconstructing $w = a + \rho\theta$ gives a physical invariant probability concentrated on $\{\mathcal{K}(w) > 0\}$. Choose an ergodic component. It too has full mass in this set, which is disjoint from $\mathcal{S}$.

For the bounded continuous function $F(w) = \min\{1, \|(I - P_x)w\|_2^2\}$, its integral under this component is strictly positive. The stationary ergodic theorem implies, by disintegration with respect to the initial state, that for almost every deterministic initial state relative to this component the asserted time-average limit holds with probability one. Undoing the amplitude and time normalization preserves the sign of $\mathcal{K}$, the shear subspace, ergodicity, and positivity of this time-average criterion (with the original-amplitude version of F). This proves Theorem 1.1.

8 Scope and checks

- The result concerns a fixed elongated aspect ratio $0 < \varepsilon < 1$ and single-mode vorticity forcing $\sigma \sin y \, dW$. It is equivalent to horizontal velocity forcing $\sigma \cos y \, dW$, up to the harmless Brownian sign convention.
- The threshold is nonquantitative. The argument does not establish the draft's particular joint scaling $\nu = o(\varepsilon^{2/3})$ as $\varepsilon \downarrow 0$.
- It proves the existential long-time conclusion and a non-shear ergodic invariant measure. It does not prove escape for every small perturbation or a positive transverse Lyapunov exponent. In particular, the original statement quantifying over *every* state near the shear subspace is false: for an exact shear, $F(\omega_t) = 0$ for all t, and the proposed exit time is infinite.
- The exact stochastic cancellation uses forcing in the eigenspace $A = 1$. General shear forcing with several vertical frequencies requires a different argument.
- At $\varepsilon \geq 1$, the cone $\{\mathcal{K} > 0\}$ is empty. The proof therefore cannot erroneously imply the same conclusion on shorter domains.
- The whole eigenspace $A = 1$ is removed in the lift, so $\varepsilon^{-1} \in \mathbb{N}$ causes no coercivity gap.
- The zero-viscosity limit is taken only for invariant *boundary* laws whose normalized directions already satisfy the compactness estimate. No interchange of an uncontrolled long-time limit with a Galerkin or inviscid limit is used.
- The frozen scalar inviscid random system can still have zero Lyapunov exponent at fixed noise correlation time. The present proof uses the viscous identity before taking the limit; its normalization would lose tightness in that purely inviscid model.

References

[1] M. Hairer and J. C. Mattingly, *Ergodicity of the 2D Navier–Stokes equations with degenerate stochastic forcing*, Ann. Math. 164 (2006), 993–1032. https://arxiv.org/abs/math/0406087. Background for the Markov and parabolic framework; the uniqueness hypotheses of that work are not invoked here.

[2] Z. Lin and M. Xu, *Metastability of Kolmogorov flows and inviscid damping of shear flows*. https://arxiv.org/abs/1707.00278. Background on Hamiltonian methods for shear linearizations; Lemma 5.1 is proved directly in this note.